

PURU JAIN

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EDUCATION

Georgia Institute of Technology

B.S. Computer Engineering — GPA: 3.90/4.00

Atlanta, GA

Expected May 2028

- **Coursework:** Data Structures & Algorithms, Computer Architecture, Objects & Design, Linear Algebra, Probability & Statistics.

TECHNICAL SKILLS

Languages: C++, Python, Go, TypeScript, JavaScript, Java, SQL, MATLAB, Bash

Backend & Distributed Systems: PostgreSQL, PostGIS, Apache Kafka, RocksDB, gRPC, REST, Debezium, Go-Chi, FAISS, pgvector, Docker, Git, React Native

ML Systems: PyTorch, CUDA, Triton, SpikingJelly, Ray Tune, SLURM, DDP/FSDP, vLLM, TensorRT-LLM, quantization (SmoothQuant/GPTQ/AWQ)

Security & Systems: Binary reverse engineering, exploit development, Linux hardening, AFL++, ASAN/UBSAN, angr, Rust

Robotics & Hardware: ROS, PyMavLink, OpenCV, Altium, SOLIDWORKS, PCB design, Verasonics phased arrays

EXPERIENCE

Founder & Distributed Systems Lead — KickBack

2025 – Present

Social Platform

Atlanta, GA

- Architected the backend as 12 bounded microservices (Auth, Social Graph, Feed, Plans, Notifications, etc.) in **Go**, using Go-Chi with decoupled REST/gRPC contracts for independent deployability and versioned service boundaries.
- Designed relational data models in **PostgreSQL** backing the friend graph and plan lifecycle, enforcing cross-cutting privacy constraints and block/report semantics on every read path.
- Integrated geospatial indexing (**PostGIS**, Geohash, H3) and vector similarity search (**FAISS**, **pgvector**) to power low-latency location- and interest-based recommendation pipelines.
- Built a cross-platform **React Native**/TypeScript client with OAuth (Google/Apple), JWT refresh-token rotation, and device-scoped push notifications.
- Led a cross-functional engineering, design, and growth team through agile sprint planning, code review, and release cycles.

ML Systems Researcher — Neuromorphic Computing (HiAER-Spike)

2025 – Present

UC San Diego

San Diego, CA

- Led memory-efficient training of Spiking Neural Networks on the DVS Gesture dataset in **PyTorch**/SpikingJelly using surrogate-gradient descent for direct event-stream learning.
- Engineered hardware-constrained ANN-to-SNN conversion and compression pipelines for CIFAR-10, folding learned parameters into hardware thresholds and applying quantization for fixed-budget accelerator deployment.
- Orchestrated distributed hyperparameter search across GPU clusters (**SLURM** + **Ray Tune**), scaling experiment throughput to map accuracy/efficiency frontiers under strict parameter budgets.
- Investigated numerical fidelity and missing-spike divergence across the software/hardware boundary, deploying bitstreams to the neuromorphic fabric via the Neuroscience Gateway (NSG) to validate simulator correctness against hardware execution.

Undergraduate Research Assistant — Ultrasound Beamforming

Aug 2025 – Present

Wallace H. Coulter Dept. of Biomedical Engineering, Georgia Tech

Atlanta, GA

- Implemented phase-conjugation beamforming in **MATLAB** using Hilbert transforms and complex-conjugate TX-element Green's-function weighting to correct wavefront aberration through organic structures.
- Achieved **6–15 dB** focal gain, **30–70%** sidelobe suppression, and **1–5 mm** focal-localization improvement over steered delay-and-sum under 30–60% aperture blockage.
- Modeled curvilinear phased-array geometry in CAD for intercostal ultrasound access under Verasonics hardware constraints.

Low-Voltage Electrical Engineer — HyTech Racing

Jan 2026 – Present

Georgia Institute of Technology

Atlanta, GA

- Designed racecar monitoring and regulation PCBs in **Altium**, applying AC/DC circuit analysis and interpreting component datasheets for reliable low-voltage subsystems.

Research Intern

May 2023 – Jun 2023

UCLA Dept. of Molecular, Cell, and Developmental Biology

Los Angeles, CA

- Annotated gene functions with GENSAS and BLASTP; corrected gene-identification algorithm errors to improve annotation accuracy by 3%.

PROJECTS

Ethical Hacking CTFs | *Team Co-Captain* | *Binary RE, Web Security, Exploit Dev*

Aug 2021 – Sep 2025

- Reverse-engineered 100+ challenge binaries and codebases to audit control flow, data validation, and trust boundaries using LLVM sanitizers and symbolic execution.
- Identified and exploited 150+ vulnerabilities spanning XSS, SQL injection, CSRF, SSRF, and memory-safety classes; chained 2–4 vulnerabilities in complex challenges.
- Wrote 50+ proof-of-concept exploits and remediation tests, hardening input sanitization, authorization, and protocol validation for vulnerable services.

Autonomous Underwater Vehicle (RoboSub) | *Founder, Software & Mechanical Lead*

Aug 2021 – Aug 2024

- Built a real-time **C++/OpenCV** perception pipeline (HSV thresholding, Gaussian filtering, morphological ops, contour filtering) detecting underwater targets at a sustained **15 FPS** on embedded edge compute.
- Integrated **ROS** and PyMavLink for asynchronous camera/navigation telemetry fusion at 15 Hz, with rosbag logging for replay-driven root-cause analysis.
- Authored a 20-page design history file covering system integration, root-cause analysis of 13+ failure modes, design changes, and validation results.

LEADERSHIP & AWARDS

CyberPatriot | *Team Captain*

Aug 2021 – Aug 2024

- National Qualifier (2022) and Semifinalist (2023, 2024); authored Bash hardening scripts remediating Linux privilege, service, and permission misconfigurations.

UC Berkeley Bioengineering Conference | *Team Captain*

Dec 2023 – Mar 2024

- Placed 2nd (2024) and 3rd (2023) among 200+ projects; led technical design presentations for therapeutic and ultrasound-device concepts.